

Exponential Equations

$$9^x = 27$$

$$9^3 = 81, 9^1 = 9$$

$$\text{so: } 1 < x < 2$$

$$27 = (3)(9)$$

$$\Rightarrow 3 = 9^{\frac{1}{2}} = \sqrt{9}$$

$$\Rightarrow 9 = 9^1$$

$$\text{then: } \Rightarrow (9)(9^{\frac{1}{2}}) = (9)^{1+\frac{1}{2}} = (9)^{\frac{3}{2}} = 27$$

$$\boxed{x = 1.5}$$

Day 4

